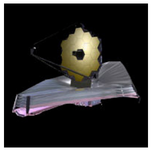

8204 - Give me a break: the search for stars in a prototypical Little Red Dot

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	Plan25-09-19-main_filler-a	NIRSpec MultiObject Spectroscopy	(1) main_onmask_star_fillerv6culled

ABSTRACT

We propose a search for stellar absorption in the Balmer break of the brightest known `Little Red Dot'. Despite two years of intense study with JWST, we still do not know whether these abundant and compact red sources are extraordinarily dense stellar systems, or red active galactic nuclei. A clean test will be provided by detecting (or not) Balmer absorption around the break. This is the only source bright enough to do so unambiguously, and so stands to teach us the nature of these new enigmatic objects.

OBSERVING DESCRIPTION

We propose deep G235M/170LP+G395M/290LP spectroscopy of the brightest known spectroscopically confirmed Little Red Dot (LRD), with F444W=22 AB mag. Our primary goals are (1) to determine if we can detect stellar absorption from the prominent Balmer break and (2) to measure associated narrow absorption in both Ha and Hb, to determine the covering fraction and density of the absorbing clouds.

We choose medium resolution gratings, which are very well-matched to the broad line-widths of the emission lines and are required to resolve stellar absorption features if present. Our exposure times (G235M/170LP = 10hr; G395M/290LP=3hr) are tuned to ensure robust detection of the CaK absorption line even in the presence of possibly significant contamination from an AGN continuum.

We will use the Multi-Shutter Array to increase the legacy value of the observations, and have other known LRDs and galaxies at comparable redshifts from the UNCOVER field as filler.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	main_onmask_star_fillerv6culled	RA: 00 14 18.6616 (3.5777567d) Dec: -30 22 58.95 (-30.38304d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 8204, Observation 1: Plan25-09-19-main_filler-a Wed Oct 08 19:00:19 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	main_onmask_star_fillerv6culled	RA: 00 14 18.6616 (3.5777567d) Dec: -30 22 58.95 (-30.38304d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: F110W; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	Main (8 sources)	Filler (2652 sources)	jwst-nirspec-hr	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	34061	3.584418	-30.383041	23.0651097422126	1	50718	3.537494	-30.359413	22.65449100886470
1	36342	3.533846	-30.380022	21.396285067384800	1	51013	3.533237	-30.357724	21.469491964478300
1	38986	3.531739	-30.375955	22.383214443919200	1	55704	3.532821	-30.348996	22.119505149090600
1	45375	3.543593	-30.373849	22.74629503412410	1	57900	3.565905	-30.344510	21.905372308261700

Proposal 8204 - Observation 1 - Give me a break: the search for stars in a prototypical Little Red Dot

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G235M/F170LP)	c1-withskyb	3 Shutter Slitlet	3.5573683333333332 Degrees -30.367622222222224 Degrees	182.4029076146386		-0.1	3	12	17681.735
2	1 (G235M/F170LP)	c1-withskyb	3 Shutter Slitlet	3.5573683333333332 Degrees -30.367622222222224 Degrees	182.40290837673206		0.1	3	12	17681.735
3	2 (G395M/F290LP)	c1-withskyb	3 Shutter Slitlet	3.5573683333333332 Degrees -30.367622222222224 Degrees	182.4029076146386		-0.1	3	3	4420.434
4	2 (G395M/F290LP)	c1-withskyb	3 Shutter Slitlet	3.5573683333333332 Degrees -30.367622222222224 Degrees	182.40290837673206		0.1	3	3	4420.434
5	3 (PRISM/CLEAR)	c1-withskyb	3 Shutter Slitlet	3.5573683333333332 Degrees -30.367622222222224 Degrees	182.40290799568766			3	3	2232.1

Special Requirements

MSA Scheduled Aperture PA 182.3926 to 182.3926 Degrees (V3 43.818077 to 43.818077)